

## Activity 2 — 5 Whys Root Cause Analysis

### Meridian Health Screening — the 07:00 blood-test backlog

|                  |                                                                                                   |
|------------------|---------------------------------------------------------------------------------------------------|
| <b>Topic</b>     | Topic 1                                                                                           |
| <b>Objective</b> | A2 · K2, K5 — Identify root causes with team members using structured group facilitation.         |
| <b>Duration</b>  | 45 minutes                                                                                        |
| <b>Grouping</b>  | Teams of 3-5                                                                                      |
| <b>Tools</b>     | 5 Whys ed-tool, ChatGPT / Copilot / Gemini                                                        |
| <b>Ed-tool</b>   | 5 Whys Tool — <a href="https://alfredang.github.io/5whys/">https://alfredang.github.io/5whys/</a> |

### The Scenario

Meridian Health operates six health-screening centres in Singapore. Corporate clients book annual screening packages; the fasting blood draw must happen between 07:00 and 09:30.

Over the last four months, 31% of morning appointments overrun by more than 40 minutes. Corporate clients have escalated twice. One major client (1,400 employees, S\$310,000 annual contract) has given notice of review. The Operations Manager's proposal is to open a seventh centre at a capital cost of S\$1.2m.

The Clinical Director is unconvinced. She has asked your team to establish the ROOT CAUSE before the board approves any capital expenditure. The nurses tell you the same thing every morning: "We're just short-staffed." The data below says something more interesting.

### The Evidence

*Meridian Health — morning screening operations, 4-month sample*

| Observation                                        | Value         |
|----------------------------------------------------|---------------|
| <b>Booked slots 07:00-09:30</b>                    | 48 per centre |
| <b>Phlebotomists rostered 07:00</b>                | 3 of 5        |
| <b>Phlebotomists rostered 08:30</b>                | 5 of 5        |
| <b>Mean draw time, trained staff</b>               | 6 min         |
| <b>Mean draw time, relief staff</b>                | 14 min        |
| <b>Relief staff share of morning roster</b>        | 40%           |
| <b>Repeat draws (failed first attempt)</b>         | 18%           |
| <b>Clients arriving without fasting compliance</b> | 12%           |
| <b>Lab courier pickup (fixed)</b>                  | 09:45         |
| <b>Samples missing the 09:45 courier</b>           | 23%           |
| <b>Permanent staff turnover, 12 months</b>         | 44%           |

## Discussion Questions

1. Run the 5 Whys to five levels. At which level did you stop describing WHAT happens and start explaining WHY the system produces it?
2. The nurses say 'we're short-staffed'. The data shows all 5 phlebotomists are rostered by 08:30. What is the real constraint — headcount, or something else?
3. Trace the causal chain to 44% annual turnover. If turnover is the root cause, what does that say about the S\$1.2m seventh centre proposal?
4. Where could this 5 Whys chain have gone wrong? Identify one point where a different, equally plausible 'why' would have led you somewhere completely different.
5. The GenAI facilitator challenged at least one of your answers as opinion rather than fact. Which one, and were you able to supply the evidence?

## Step-by-Step

6. Read the Meridian Health scenario and study the operations table.
7. Appoint a facilitator (asks the whys and keeps the team on one branch) and a scribe.
8. Open the 5 Whys ed-tool at <https://alfredang.github.io/5whys/>
9. Enter the problem statement in the tool's problem field, using numbers from the table.
10. Ask Why #1. The facilitator's job: reject any answer that is an opinion, and ask 'what evidence supports that?'
11. Continue to five levels, entering each why and answer in the tool. Stay on ONE causal branch.
12. Mark the level where the answer stopped being a description of events and became a system explanation.
13. Run the same problem through the GenAI 5 Whys prompt from the slide, answering as the clinical team would.
14. Compare the two chains. Where they diverge, decide which branch the EVIDENCE supports and record why.
15. Write your root cause statement in one sentence, and note which levels remain unverified.
16. Present to the room in 90 seconds: the chain, the root cause, and your verdict on the S\$1.2m proposal.

## The GenAI Prompt

Act as a root cause analysis expert facilitating a clinical operations team.

Problem: 31% of morning health-screening appointments at Meridian Health overrun by more than 40 minutes, and 23% of blood samples miss the fixed 09:45 lab courier.

Evidence: only 3 of 5 phlebotomists are rostered at 07:00 (all 5 by 08:30); relief staff take 14 min per draw vs 6 min for trained staff; relief staff are 40% of the morning roster; 18% of draws need a repeat attempt; permanent staff turnover is 44% a year.

Apply the 5 Whys technique. Ask ONE why at a time and wait for my answer before asking the next. Go five levels deep.

At each level, state what EVIDENCE would confirm or refute that level – do not accept my answer at face value. If my answer is an opinion rather than a fact, say so and ask me for the data that would support it.

After level 5, summarise: the root cause, the evidence supporting it, and which levels remain unverified assumptions.

## Trainer Debrief

*Expected answers and the points to draw out. Trainer reference.*

EXPECTED CHAIN (teams may reach this by slightly different routes — accept any chain that is evidence-supported and reaches a systemic cause):

Why 1: Appointments overrun -> draws take longer than the slot allows.

Why 2: Draws take longer -> 40% of morning staff are relief staff at 14 min vs 6 min, and 18% of draws need a repeat.

Why 3: So many relief staff -> only 3 of 5 permanent phlebotomists are rostered at 07:00.

Why 4: Under-rostered at 07:00 -> permanent staff turnover is 44%/yr; vacancies are backfilled with relief staff who are not trained to the centre's protocol.

Why 5: Turnover is 44% -> the early shift is unattractive and there is no structured onboarding or retention path for phlebotomists.

ROOT CAUSE: a workforce retention and onboarding failure, surfacing as a capacity problem.

### KEY TEACHING POINTS:

1. The S\$1.2m seventh centre solves the SYMPTOM. A new centre staffed by the same 40% untrained relief pool reproduces the same overrun — you buy the problem twice.
2. 'We're short-staffed' was the team's stated cause and it was WRONG in the way that matters: they are not short of headcount, they are short of TRAINED headcount at 07:00. This distinction is worth S\$1.2m.
3. Facilitation discipline: one why at a time, and every level must survive 'what evidence supports this?'. The GenAI acting as challenger models the facilitator behaviour learners should copy — this is the K5 group facilitation outcome.
4. Note the branch point: at Why 2 a team could have followed the 12% fasting non-compliance instead and landed on 'client communication'. That is a legitimate SECOND cause. 5 Whys gives depth on one branch — which is exactly why Activity 3 uses Fishbone for breadth.

## Self-Check — Is Your Output Finished?

Your root cause is sound when you can answer YES to all three: (1) If this cause were removed, would the 31% overrun stop recurring — not just improve this month? (2) Is every level in the chain supported by a number in the evidence table, or explicitly flagged as an assumption? (3) Does the root cause point at a SYSTEM (policy, process, incentive) rather than at a person? If your chain ends at 'the relief staff are slow', you have found a symptom and blamed a human — go two levels deeper.